

DELHIVERY NETWORK INTELLIGENCE STRATEGY

Optimizing Delivery ETAs with Graph-Based Network Intelligence

A data-driven roadmap to reduce SLA breaches, improve ETA accuracy, and prioritize hub & corridor investment across Delhivery's national network.

The Business Problem

ETA inaccuracy is a network-wide issue with concentrated root causes

Why ETA Accuracy Matters

- ETAs are the customer's primary trust signal — every missed promise drives support escalations and erodes confidence in Delhivery's service.
- Late deliveries cascade into reactive re-planning, overtime, and emergency capacity decisions across operations teams.
- Today's routing-engine (OSRM) ETA is accurate within 15% for only 3.8% of deliveries — meaning the current ETA promise is wrong over 96% of the time.
- This is solvable: delay is concentrated in identifiable hubs and corridors, not spread evenly across the network.

Project Objective

Build a graph-based intelligence system that (1) improves ETA prediction beyond OSRM, (2) identifies bottleneck hubs and delayed corridors, (3) creates an FTL vs Carting decision framework, and (4) produces a quantified, prioritized investment roadmap.

1,657

Facilities Analyzed

2,783

Active Corridors

144,867

Trip Segments

3.8% → 61.3%

Share of ETAs within 15% of actual: OSRM baseline vs. graph-enhanced model

Network Overview

Delhivery's facility network modeled as a directed corridor graph

Graph Structure

- Nodes = Facilities. Each of the 1,657 source/destination centers is a node in the network.
- Edges = Corridors. Each directed source → destination pair is an edge carrying delay, volume, and route-type attributes.
- Main network component: 1,353 facilities (81.7%) and 2,463 corridors (88.5%) form one connected core — this is where centrality and embeddings are computed.
- 64 separate sub-networks exist outside the main component, representing isolated or low-connectivity regional routes.

1,657

Total Facilities (Nodes)

2,783

Total Corridors (Edges)

81.7%

Facilities in Main Connected Network

88.5%

Corridors in Main Connected Network

64

Disconnected Sub-Networks

93.4%

Corridors Exceeding 20% Delay

Recommended visual: Interactive directed network graph (NetworkX / pyvis) — nodes sized by traffic volume, colored by structural risk score, edges colored red where median delay exceeds 20%.

The Current ETA Challenge

OSRM-based routing estimates systematically underestimate real-world transit time

OSRM Limitations

- OSRM assumes free-flow road conditions and does not account for hub dwell time, loading/unloading, or real congestion.
- Average actual transit time runs at ~2.25x the OSRM estimate across the network.
- Carting routes average 2.67x OSRM time; FTL routes average 2.06x — both route types are systematically underestimated.
- 85–91% of segments exceed their OSRM time estimate — this is a structural mismatch, not noise.

Delay Patterns

2.25x

Avg. Actual vs. OSRM Time Ratio

91.1%

FTL Segments Exceeding OSRM Time

85.3%

Carting Segments Exceeding OSRM Time

2,598

Corridors with >20% Median Delay (of 2,783)

Bottom line: under the current OSRM-based ETA approach, customers are told a number that is wrong by more than 15% over 96% of the time.

Graph Intelligence Framework

How network science turns facility data into operational insight

1

Nodes & Edges

Every facility becomes a node; every shipping corridor becomes a directed edge carrying real operational data — delay ratio, volume, and route type. This turns a spreadsheet of trips into a map of how the network actually behaves.

2

Centrality Scores

Algorithms identify which facilities sit on the most critical paths (betweenness), which are most connected (degree), and which have the most network-wide influence (PageRank) — surfacing hidden chokepoints invisible in simple volume reports.

3

Network Embeddings

Node2Vec encodes each facility's position in the network as a numerical fingerprint. Two facilities with identical distance and traffic can have very different embeddings if one sits in a congested cluster and the other doesn't — this fingerprint feeds directly into ETA prediction.

In plain terms: this framework lets us answer 'which facility, if it slows down, will hurt the most other deliveries?' — a question raw volume data cannot answer.

Bottleneck Hub Analysis

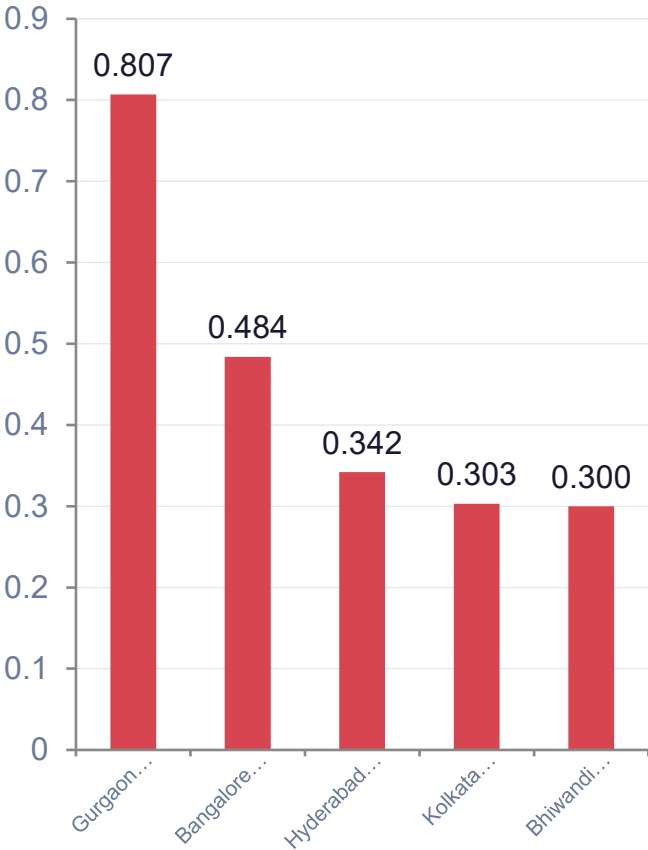
Five facilities account for over 62% of all SLA-breach segments network-wide

Rank	Facility	Structural Risk Score	Betweenness Centrality	SLA Breach %
1	Gurgaon Bilaspur HB (Haryana)	0.807	0.330	25.9%
2	Bangalore Nelamangala H (Karnataka)	0.484	0.187	14.6%
3	Hyderabad Shamshabad H (Telangana)	0.342	0.139	6.1%
4	Kolkata Dankuni HB (West Bengal)	0.303	0.136	5.3%
5	Bhiwandi Mankoli HB (Maharashtra)	0.300	0.104	10.6%

Operational Implications

- Gurgaon Bilaspur HB has nearly 2x the betweenness centrality of any other facility — a true single point of failure for North Indian network throughput.
- 4 of the top 5 hubs rank in the top 10 for SLA-breach contribution — structural importance and operational delay reinforce each other.
- These 5 facilities represent <0.4% of all nodes but drive the majority of network-wide SLA risk — a clear 80/20 investment case.

Top 5 Bottleneck Hubs
— Risk Score



Corridor Delay Audit

93.4% of corridors exceed the 20% delay threshold — two distinct delay profiles emerge

Worst Corridors by Total Delay Impact

Corridor	Median Delay	Trips	Impact (min)
Shirpur → Dhule MIDC (MH)	21.2x	14	9,338
Hazrat Jandaha → Muzaffarpur (BH)	11.4x	14	6,893
Benipatti → Muzaffarpur (BH)	10.0x	15	6,789
Simlapal → Hura Central (WB)	14.2x	8	5,502
Helencha → Kolkata Dankuni HB (WB)	15.4x	5	5,299

These extreme-delay, low-volume corridors point to OSRM road-network data quality issues for regional/rural routes — not capacity problems.

High-Volume Hub Corridors Above 20% Delay

- Bangalore Nelamangala → Bengaluru Bommasandra: 151 trips, 1.42x median delay
- Bhiwandi Mankoli HB → Mumbai Hub: 105 trips, 2.18x median delay

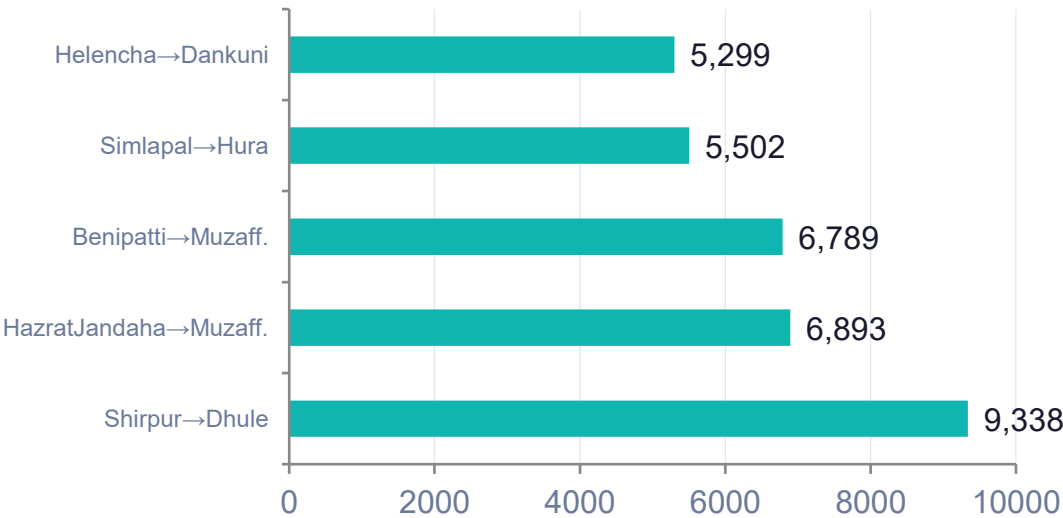
2,598

of 2,783 corridors exceed +20% delay (93.4%)

129,351

Total SLA-breach segments network-wide

Top 5 Corridors — Total Delay Impact (min)



Model Performance Comparison

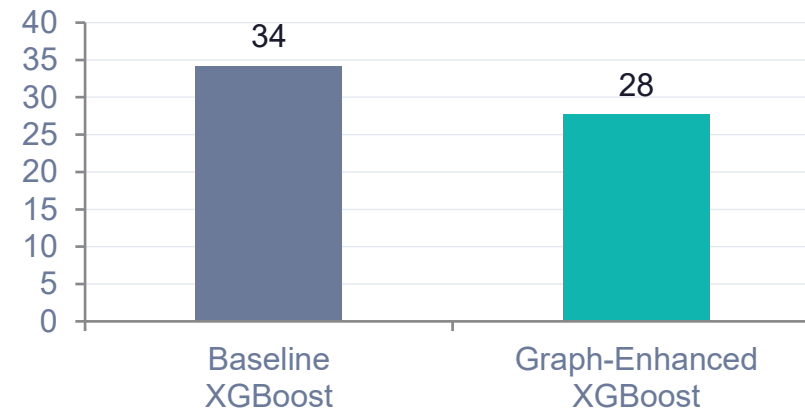
Graph-enhanced ETA model cuts prediction error by 19% versus baseline ML

Model	MAE (min)	RMSE (min)	R ²	% Within 15% of Actual ETA
OSRM Estimate (current state)	204.0	366.9	0.624	3.8%
Random Forest (baseline ML)	35.4	69.0	0.987	56.9%
XGBoost (baseline ML)	34.2	63.1	0.989	56.1%
XGBoost + Graph Embeddings (recommended)	27.7	50.9	0.993	61.3%

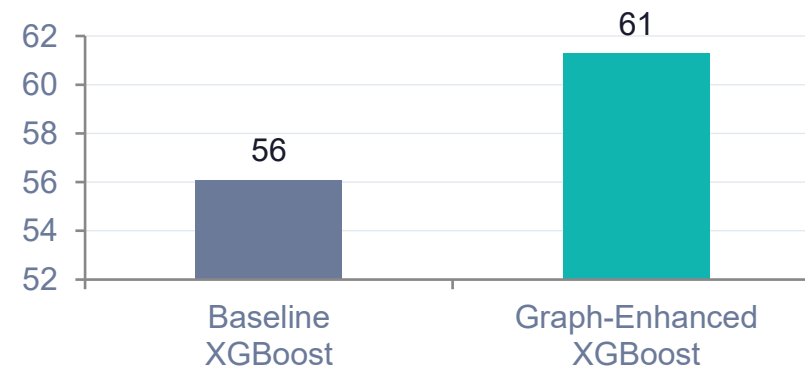
The Graph Advantage

- Graph embeddings encode each facility's network position — a signal conventional features (distance, time-of-day) cannot capture.
- Same hyperparameters as the baseline model — the entire 19.1% MAE improvement is attributable to network-structure features alone.
- Drop-in upgrade: no change to physical operations required to capture this gain — purely a model deployment decision.

Mean Absolute Error (minutes)



% Predictions Within 15% of Actual ETA



FTL vs Carting Decision Framework

99% of corridors already use the optimal route type — 14 corridors flagged for review

Key Findings

- A recommendation model evaluates corridor volume, distance, and delay profile to suggest FTL or Carting — validated at 99% classification accuracy.
- 99.5% of corridors (2,769 of 2,783) already use the model-recommended route type — confirming current allocation logic is largely sound.
- 14 corridors are flagged as mismatched, representing an estimated 53 hours of recoverable cumulative transit time.
- 3 high-volume corridors account for ~46 of the 53 hours — the recovery opportunity is highly concentrated.

Recommended Usage Patterns

Route Type	# Corridors	Avg. Distance	Avg. Volume
FTL	1,723	25.6 km	8.2 trips
Carting	1,060	11.6 km	11.6 trips

Decision Matrix — Current vs. Recommended

Current → Recommended	FTL	Carting
Currently FTL	1,723	0
Currently Carting	14	1,046

Top Recovery Opportunities (Carting → FTL)

Corridor	Trips	Min. Saved
Bangalore Nelamangala ↔ Bengaluru Bommasandra	127 / 102	1,065 / 846
Delhi Gateway HB → Gurgaon Bilaspur HB	100	833

Business implication: this is a low-risk, high-confidence fine-tuning opportunity — not a structural overhaul. Re-run the model quarterly as volumes evolve.

Business Impact Simulation

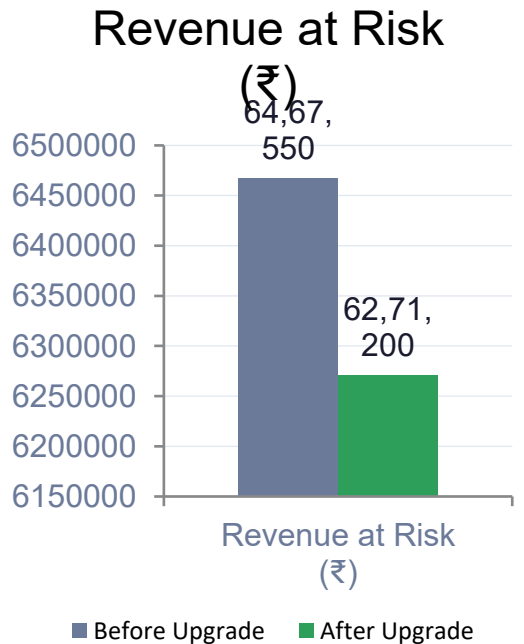
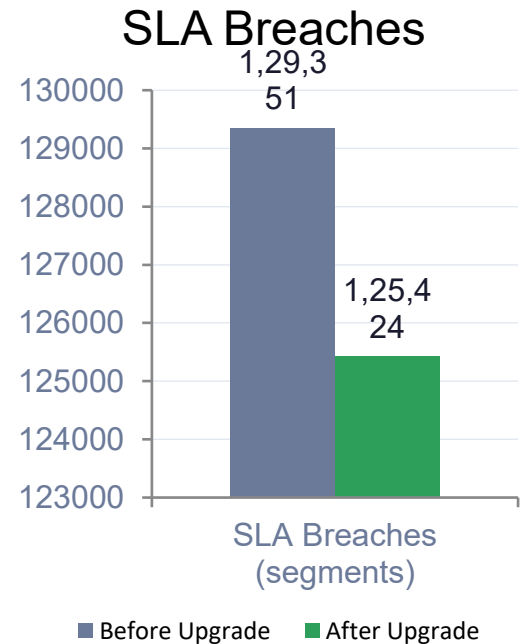
Upgrading just 3 hubs (25% of network volume) reduces SLA breaches by 3.0%

Simulation Assumptions

- Upgrade the top 3 bottleneck hubs: Gurgaon Bilaspur HB, Bangalore Nelamangala H, Hyderabad Shamshabad H.
- Conservative assumption: a 20% reduction in outbound delay ratio at each upgraded hub (typical for added sorting capacity / dock doors).
- Revenue-at-risk valued at an illustrative ₹50 per SLA-breach segment
- Effect modeled only on segments directly originating from the 3 hubs — downstream ripple effects are NOT included, making this a conservative floor estimate.

36,662 segments affected — 25.3% of total network volume — from upgrading just 3 of 1,657 facilities.

Metric	Before Upgrade	After Upgrade	Change
SLA Breaches (segments)	129,351	125,424	-3,927 (-3.0%)
Revenue at Risk (₹, illustrative)	6,467,550	6,271,200	+₹196,350 recovered



Strategic Recommendations

A prioritized, three-horizon roadmap tied directly to the findings above

IMMEDIATE

0–3 months

- Capacity & process review at Gurgaon Bilaspur HB and Bangalore Nelamangala H — highest structural risk scores and SLA-breach contributions
- Deploy graph-enhanced ETA model to production — 19.1% MAE improvement, no operational change required

MEDIUM-TERM

3–6 months

- Regional cluster review: Bhiwandi Mankoli HB + Mumbai/Pune corridors; Kolkata Dankuni HB + Eastern feeder corridors
- Re-assign route type on the 3 highest-impact flagged corridors — captures ~46 of 53 recoverable hours

LONG-TERM

6–12 months

- OSRM map-data correction audit for extreme-outlier rural/regional corridors — improves ETA accuracy and model inputs network-wide
- Preventive capacity planning at Hyderabad Shamshabad H ahead of projected volume growth

Final Executive Takeaways

ETA unreliability at Delhivery is concentrated, measurable, and addressable — not a diffuse problem requiring broad investment.

What Delhivery Should Do Next

- Deploy the graph-enhanced ETA model immediately — a low-risk software upgrade with a 19.1% accuracy gain
- Launch capacity reviews at Gurgaon Bilaspur HB and Bangalore Nelamangala H first — the two highest-leverage interventions identified
- Establish the corridor-monitoring dashboard to track all 2,598 high-delay corridors and catch new ones proactively
- Re-run the FTL/Carting and bottleneck analyses quarterly as the network evolves

Expected Business Value

19.1%

Reduction in ETA prediction error (MAE)

+5.2pp

More deliveries with ETA accurate within 15%

-3,927

SLA breaches avoided from top-3 hub upgrades

₹196K+

Revenue at risk recovered (illustrative; recalibrate with actuals)

Thank you — we welcome the opportunity to walk through these findings in detail and support translating this analysis into an executable rollout plan.